Annals of Data Science
https://doi.org/10.1007/s40745-024-00540-5

A Deep Convolutional Neural Network-Based Approach for Visual Search & Recommendation of Grocery Products

Nawreen Anan Khandaker[1] · **Amrin Rahman**[1] · **Amrin Akter Pinky**[1] · **Tasmiah Tamzid Anannya**[1]

Received: 22 August 2023 / Revised: 8 April 2024 / Accepted: 22 April 2024
© The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature 2024

Abstract
Search and recommendation are two essential features of any e-commerce website for finding and purchasing a specific product. Visual Search is a promising and quick method in comparison to a textual-based search method. Hence, the objective of this research is to propose a conceptual framework for developing a visual search and recommendation system for grocery products using Ensemble Learning with CNN models. Traditional Deep learning and Ensemble Learning techniques were implemented with a publicly available and a self-made data set containing 3174 and 3162 images respectively. Various combinations of the suitable models found from research findings were used to find the best-fitted model for both the search and recommendation functionalities. All the models were evaluated using suitable performance metrics and the Ensemble Learning approach performed better. The best-performed results for visual searching are obtained by incorporating VGG16 and MobileNet with an accuracy of 99.8% for classification and in the case of product recommendation, the combination of MobileNET and ResNET50 performs better than other techniques.

Keywords Visual search · Recommendation · Ensemble learning · Deep learning

✉ Tasmiah Tamzid Anannya
tasmiah_anannya@cse.mist.ac.bd

Nawreen Anan Khandaker
anan.naw11@gmail.com

Amrin Rahman
amrinrahman14@gmail.com

Amrin Akter Pinky
amrinpinky3379@gmail.com

1 Department of Computer Science and Engineering, Military Institute of Science and Technology (MIST), Dhaka, Bangladesh

Published online: 23 May 2024

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

1 Introduction

The proliferation of e-commerce websites is increasing day by day due to the escalating demand of users. A user may search for a specific product on these websites through the search bar. Research shows that if a site search box is offered then 30% of visitors use it and it is 1.8 times more effective in comparison with manually searching for the product [1]. A quick and smooth search engine provides users with an efficient and less time-consuming search experience along with a higher probability of finding the target product. Search and recommendation techniques available on the most popular e-commerce websites are mainly text-based. However, text-based search is not always convenient to the users, especially in cases where the users find the product name hard to type or the product name is troublesome to perceive, or it is written in some other language which requires extra effort to search for that desired product. Search engines with low vocabulary for synonyms also fail to show the correct product. For example, pasta can also be written as spaghetti. According to [2], 70% of websites require users to search by the exact term that the website uses which fails to return relevant products when given another synonymous term. Additionally, 60% of the websites do not conduct thematic search queries, and 84% of them are unable to filter products by adjectives such as "cheap" or "quality" [2]. This cuts down some of the efficiency and convenience failing the whole purpose of having these e-commerce websites in the first place.

Now, in this big data era, due to the available optimization techniques, data mining principles, business data mining applications, and advancements in big data analytics, some studies [3–6] suggest that integrating these could significantly enhance, optimize and improve user experiences. According to these studies, data science plays a crucial role in handling large datasets which involves gathering data, categorizing them, and annotating them with relevant metadata. Data scraping, data cleaning, and tagging to ensure the quality and relevance of the dataset can easily be employed using available data science techniques. Searching time, ease of searching, convenience, ease of browsing, and spotting the desired products are the primary parameters that need to be given priority while developing these types of e-commerce websites for grocery shopping. Here comes the need to introduce visual searching of the products when the user can not find products just by depending on the given arrangement provided by the websites. A visual searching system promises to overcome the issues that users face using a textual searching system. Websites that use visual searching methods these days mostly utilise machine learning and deep learning technologies. These approaches provide the facility of constant improvement with the experiences gained from the upcoming data. Google Lens, Pinterest Lens, Amazon camera search, Snapchat, eBay, etc are some platforms that widely use visual search in their search engines.

Recommending similar types of products to consumers through e-commerce websites is also an important aspect that draws the attention of users as well as is beneficial for the website owners. Research shows that websites containing visual recommendation systems opt for approaches like content-based, knowledge-based, collaborative-based (either model or memory-based), critiquing-based, group-based and hybrid methods [7]. Some of these approaches emphasize the nearest neighbours,

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

while others go for category exploitation or keyword exploitation. Researchers have not explored much on the recommendation technique for grocery products in the current existing e-commerce websites.

Deep learning algorithms typically attain a high level of accuracy in categorizing images and thus can be utilized for practical applications while searching grocery products through images and suggesting related products. Therefore, the objective of this research is to propose a deep learning-based classification technique for grocery product searching while also proposing related products. The significant contributions that have been made to achieve the goal are listed below:

- A deep learning method incorporating different state-of-the-art techniques with several pre-trained models in a CNN architecture has been utilized for feature extraction from the input image. Then, a stacking ensemble machine learning model incorporating three traditional models has been employed for searching grocery products.
- For evaluating the performance of the proposed methodology, two machine learning techniques have been trained and tested with the same dataset for classifying grocery products which includes convolutional neural network models and ensemble learning models.
- Traditional clustering algorithms along with ensemble learning have been employed for clustering similar products and suggesting products related to searched images.
- Finally, a model is proposed utilizing various advanced deep learning techniques for implementing visual product search along with recommending similar products to e-commerce users.

The remainder of this research is organized as follows: The theoretical background and literature review are briefly introduced in Chapter 2. Chapter 3 provides a research methodology framework. Chapter 4 consists of an experimental evaluation wherein performance analysis and comparative analysis have been done of both the functionalities and representation of the proposed system. Chapter 5 concludes the research with the research outcome, contribution, limitations, and future work.

2 Related Works

Existing research focusing on the visual search engine for e-commerce websites and recommending similar products are studied, analyzed, and discussed in this section.

2.1 Visual Search of Products

A cosmetic product recognition system was implemented in [8] to recognize cosmetic products with their types, brands, and retailers to analyze a customer experience and what kind of products and brands they need. Various machine learning supervised classification methods such as Logistic Regression [9], Linear Support Vector Machine [10], Adaptive k-Nearest Neighbor [11], Artificial Neural Network [12], and Decision Tree classifiers [13] have been employed to perform the classification tasks. In [14],

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

an approach for visual query posing to perform both visual and textual queries using images was shown. Both of the works used a trained VGG16 model for the visual feature extraction [8, 14]. Another study [15] used computer vision algorithms to develop a model for creating an effective image search service.

The authors of [16] demonstrated a multimodal fusion framework that uses CNN and natural language understanding models for image-based logo recognition and brand recognition. However, authors in [17] designed their image feature recognition algorithm integrating image data mining, visual attention analysis, and feature extraction technologies to construct a novel framework for the visualized design of the product. In [18], a metric learning method for image set recognition using subspace representation was proposed where each image set is compactly represented by a subspace in a high dimensional vector space.

2.2 Visual Search and Recommendation of Products

Various works used different approaches to design their visual search engine architecture using deep learning and machine learning techniques and approach of cosine distance for image recommendation module focusing on the fashion e-retail industry [19], scanned and identified real-time world objects based on the deep learning algorithm and recommended in the last phase [20] and helped user search for a product along with its surrounding supplier through product image [21]. Large-scale visual search and recommendation algorithms were used by Alibaba [22] that deployed GoogLeNet V1 [23] and pre-trained ImageNet [24] feature, by e-commerce platform JD.com [25] that retrieved real-time image, in Microsoft Bing [26] and in Big Image Warehouse [27].

Different other search algorithms and models were also used such as Elasticsearch by [28] and [29], AlexNet [30], ZFSPPNet [31], GoogleNet [32], and ResNet [33] by authors of [26], Fuzzy Logic by [34], recommendation using Euclidean distances by [35], fine-tuning the last 10 layers of the VGG-16 network [36] by authors of [37] and YOLO 9000 [38] by [39].

The authors of [40] conducted a study on the use of different pre-trained CNN models for image feature extraction and attempted to fine-tune the selected ResNet50 model for their custom dataset. For image retrieval and recommendation, the feature similarity is measured through metrics such as Euclidean Distance, Manhattan Distance, Cosine Similarity, and Approximate Nearest Neighbors. A comparative analysis of a few of the main works symmetrical to our goal is summarized in Table 1.

The work on this subject started very recently so the amount of research done is less. Moreover, only three to four works seemed to be done showing a structured architecture of visual search and recommendation of products. The models were used separately which lacked an efficient way of choosing models. Using different combinations of these models can speed up the whole work. However, none of the previous research in this field has used the method of ensemble learning that uses a combination of multiple learning algorithms to obtain better predictive performance. Again no work has been noted specifically for a wide range of grocery products. Therefore, this study aims to develop an ensemble-based machine learning model for the searching and

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

Table 1 A comparative analysis of existing research studies

Paper	Technique	Searching	Recommendation	Machine learning	Deep learning	Ensemble learning
[8]	Visual searching approach for cosmetic product recognition	✓	×	×	✓	×
[19]	Visual search engine for e-retail	✓	✓	✓	✓	×
[20]	Visual e-commerce application using deep learning	✓	✓	×	✓	×
[22]	Large-scale visual search algorithm and system infrastructure at Alibaba	✓	×	✓	✓	×
[39]	Multimodal search engine for fashion and interior design	✓	✓	×	✓	×
Ours	Visual search and recommendation of grocery product using ensemble learning	✓	✓	✓	✓	✓

recommendation of similar grocery products along with making the system accurate and time efficient.

3 Methodology

In this study, an extended machine learning classification technique has been proposed along with the trained and tested integrating ensemble CNN models that aim to search similar images with the target product image. Random product images have been used as the input of the study from which the suggested method would determine whether similar products are available or not. Four types of techniques have been explored for visual searching and product recommendation. The framework of the methodology used in this research is illustrated in Fig. 1. The phases involved in this research are described briefly below.

3.1 Data Acquisition

Figure 1 shows the whole research methodology step by step. In the first stage, the images utilized in this study have been collected in two ways. At first, an open-source data set of grocery items is collected from Kaggle to train and test our proposed models.

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

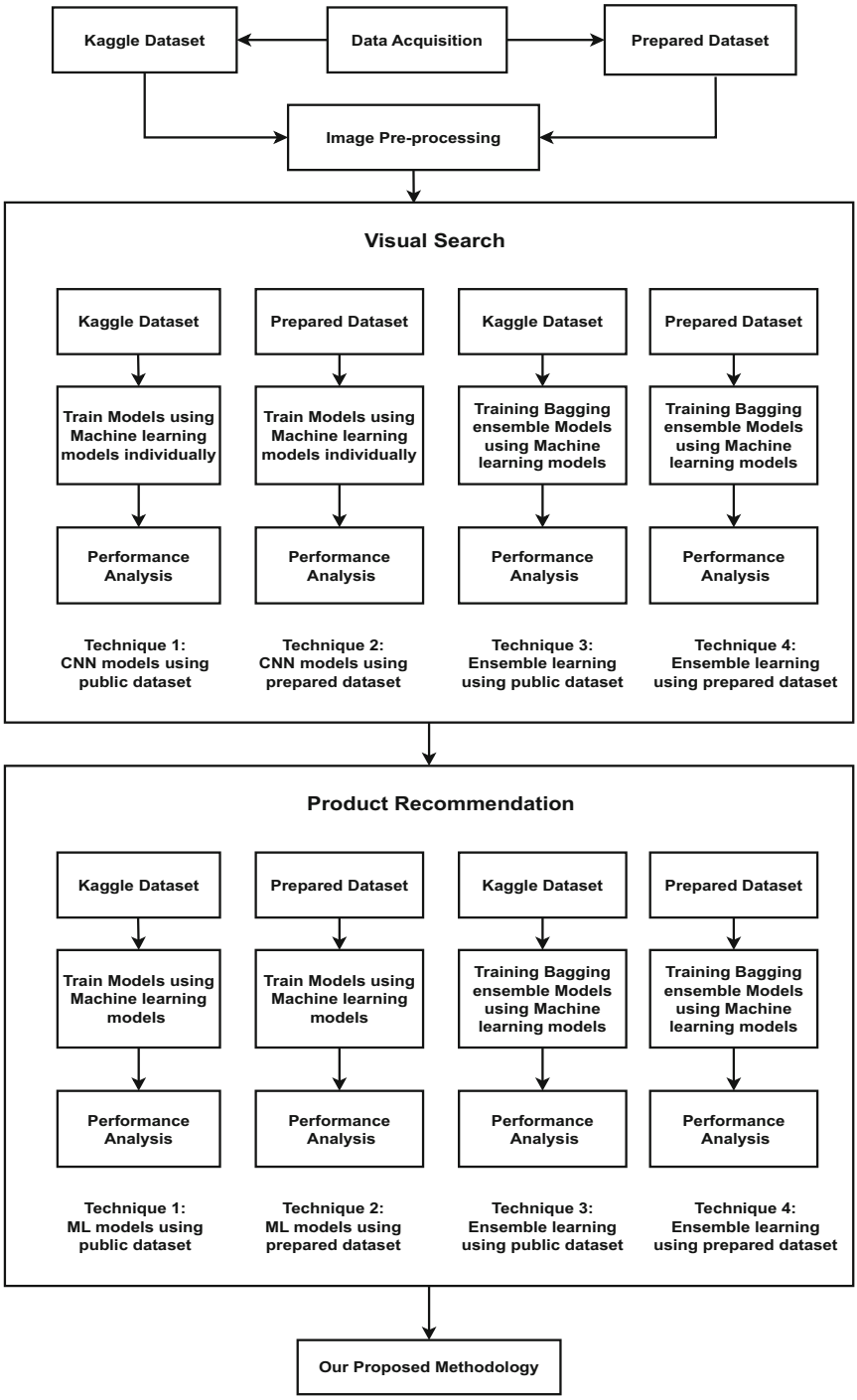

Fig. 1 Framework of visual searching and recommendation methodology

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

Fig. 2 Preprocessed image

The data set was mainly categorized into 3 groups: Fruit, Packages, and Vegetables. It contained a total of 3174 images which are further classified into 43 categories encompassing 78 sub-types. For the ease of model training and evaluation, the data set was divided into the train, test, and validation folders. We also prepared a data set having 3162 images that were divided into a total of 44 categories with three main classifications as previously collected data set from the internet. The images were captured in their actual natural habitat.

3.2 Image Preprocessing

Picture preparation for both the prepared and Kaggle dataset, two techniques from the Keras library, a high-level neural networks API built on top of Tensorflow, are used. Specifically, the ImageDataGenerator technique is used to prepare visual data for neural network training and testing, while the "reprocess input step is used to get the pictures ready for the CNN model. Preprocessing activities that the ImageData-Generator can carry out on the images include re-scaling pixel values, standardizing the image data, converting colour spaces, and applying filters to enhance particular features or remove noise which is shown in Fig. 2.

3.3 Visual Product Search Architecture

The Visual Searching Model Architecture presented in this section describes the approach for identifying grocery products in an e-commerce site. The first step is to train machine learning models for classifying different images of fruits, packages, and vegetables. For this purpose, three widely used CNN algorithms, VGG16, ResNET50, and MobileNET are chosen for the implementation of the classification and category prediction component. The research involves four types of techniques, as shown in 1. The first and second techniques involve the traditional approach of deep learning based on the Kaggle data set and our prepared data set, respectively. In both cases, the chosen models have been trained individually using the transfer learning method. In

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

the latter two strategies, various combinations of the chosen models have been trained by ensemble learning for both the Kaggle data set and the proposed data set.

3.3.1 Technique 1 and Technique 2: Deep Convolutional Neural Network-Based Approach with Kaggle Dataset and Prepared Dataset

The convolutional neural network (CNN) models VGG16, ResNet50, and MobileNet have all been pre-trained and trained on image data sets for tasks like picture categorization. Due to their performance, architectural improvements, and trade-offs between accuracy and efficiency, VGG16, ResNet50, and MobileNet are regarded as some of the best for picture categorization. Kaggle data sets and prepared data sets used in this research are divided into three groups for train, test, and validation, respectively: 80%, 10%, and 10%. These models distinguished comparable photos from the data set during training, separated them, and classified them. Test data employs the trained model and produces the expected results during testing.

3.3.2 Technique 3 and Technique 4: Ensemble Learning-Based Approach with Kaggle and Prepared Dataset

With this approach, multiple models are effectively combined to produce a single prediction system. Ensemble learning can assist in lowering variance by combining a variety of models. By combining the predictions from several models, bias can also be lessened. With the help of VGG16, ResNet50, and MobileNet, we created 4 ensemble models. The combinations of these four models are VGG-MobileNet-ResNet50, VGG16-ResNet50, VGG16-MobileNet, and MobileNet-ResNet50 respectively which is shown in Fig. 3. The identical method used for training for technique 1 was also used for ensemble learning.

3.4 Product Recommendation

The paper uses K-means clustering and K-nearest neighbour (KNN) algorithms to cluster different products and recommend products, respectively. These algorithms are considered ideal for implementing production recommendations. The models that were applied to product recognition were employed in addition to these two models. To determine the model combination that produces the best outcomes, these models have been used separately and coupled through ensemble learning.

3.4.1 Technique 1: Machine Learning Approach with Kaggle and prepared Dataset

K-Nearest Neighbors (KNN) is used in this case to recommend similar data or images. During prediction, KNN compares new examples to the stored instances depending on how similar their feature sets are, which is determined using Euclidean distance. It stores the training instances in memory. Due to the complementing advantages of both methods, combining the KNN algorithm with deep learning models frequently produces better results. To determine the best answer, a KNN combination and a few more algorithms were applied as shown in Fig. 4

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

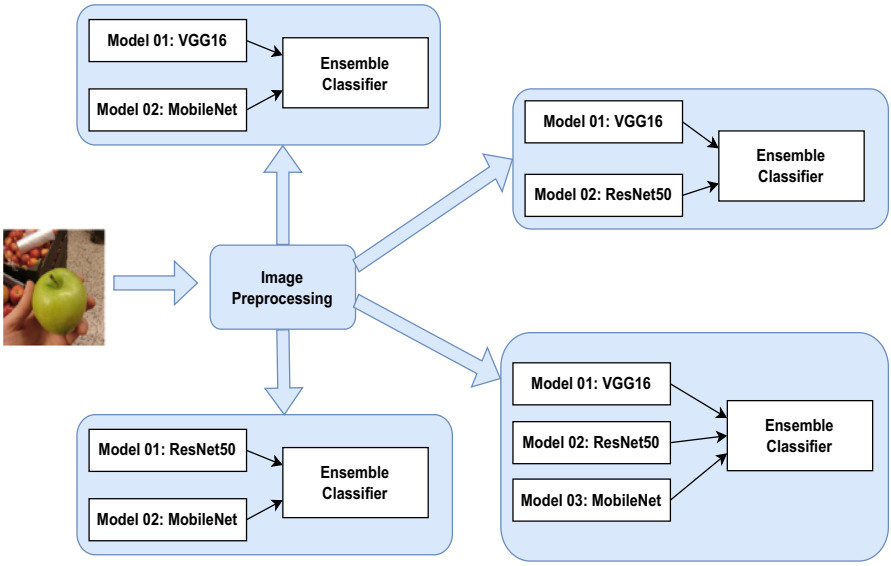

Fig. 3 Ensemble learning

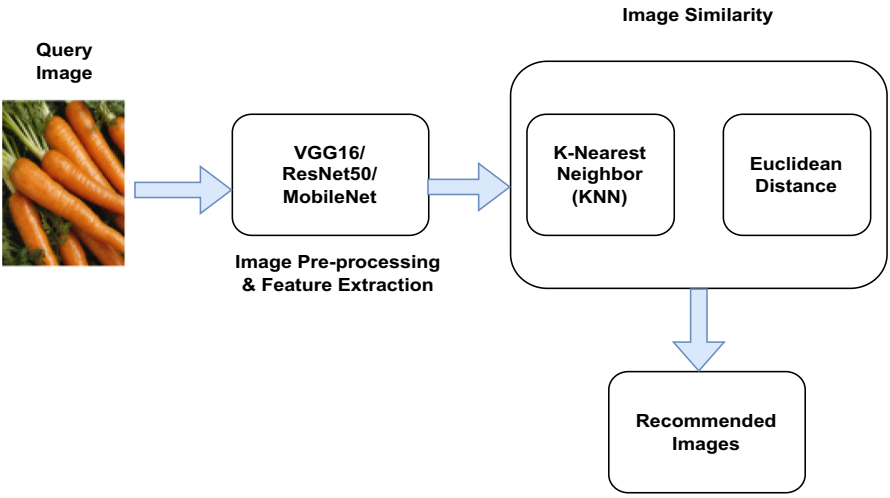

Fig. 4 Machine learning and KNN approach

3.4.2 Technique 2:Ensemble Learning Approach with Kaggle and Prepared Dataset

The chosen models are employed to construct ensemble models, which are subsequently trained. KNN with ensemble models are employed to achieve better results because ensemble learning algorithms produce results that are superior to those of individual learning algorithms. The identical strategy as before is employed here. Euclidean distance aids in locating related facts. Euclidean distance aids in locating

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

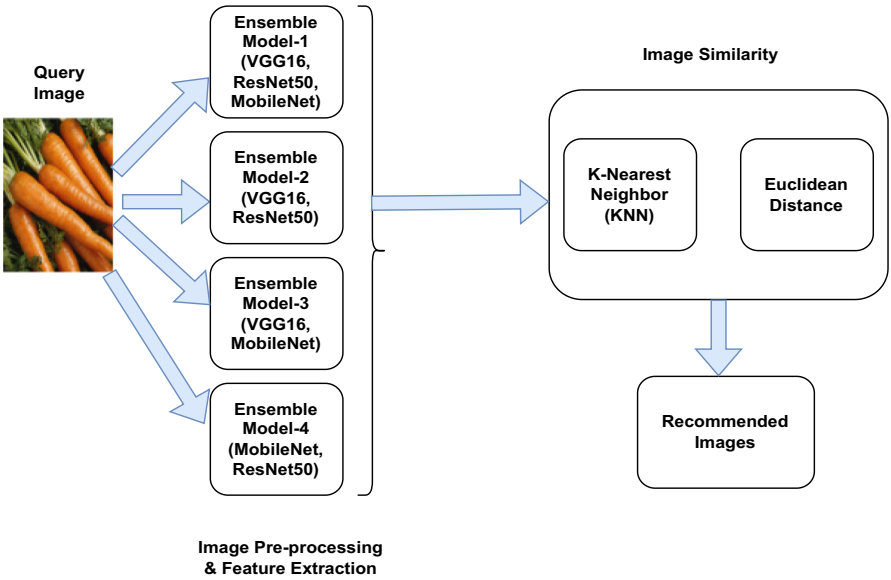

Fig. 5 Ensemble models and KNN approach

them because a recommendation for an image is required here. After performing feature extraction and image pre-processing during model training, the KNN method is applied alongside the trained models. Figure 5 depicts how image recommendations are made.

4 Proposed Model for Product Discovery and Recommendation in E-commerce

We have utilized various advanced CNN models to implement visual product search, k-means clustering and KNN to recommend similar products to e-commerce users. Based on the promising outcomes from the techniques mentioned above, we propose a model for visual searching and recommendation of e-commerce products. Figure 6 represents the detailed design architecture of the proposed system.

4.1 Visual Search

The proposed system contains two modules, the Visual Search and the Product Recommendation module. At first, the visual search module pre-processes the input image. Then the pre-processed image is fed into the classification sub-module where through the ensemble model (VGG16 and MobileNet) the category of the image is predicted. The mathematical formulation for the ensemble model that we have formed using VGG-16 and MobileNet is given below:

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

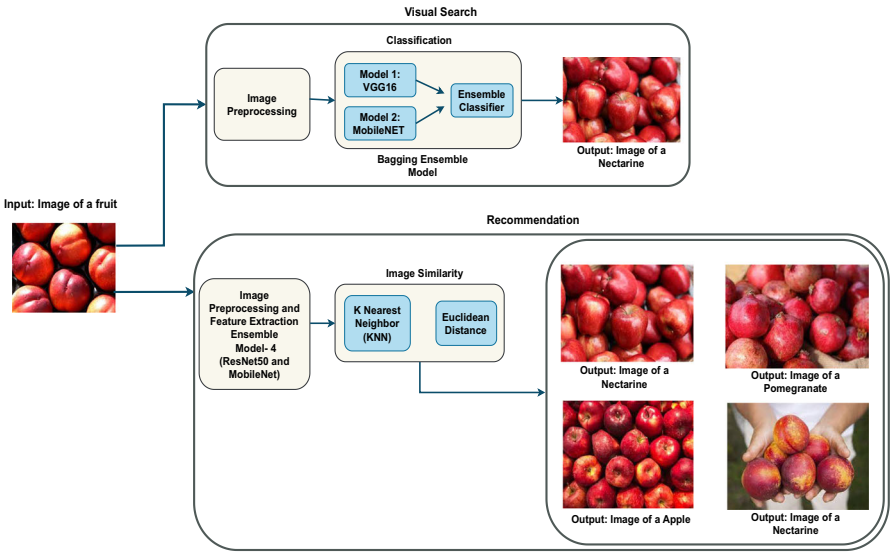

Fig. 6 System architecture of proposed system

1. For each image x_i, probability distribution across k categories from VGG-16 and MobileNet are denoted as below:

$$Y_{\text{VGG},i} = \text{VGG-16}(x_i), \quad Y_{\text{MobileNet},i} = \text{MobileNet}(x_i) \tag{1}$$

The probability distribution for model VGG-16(x_i) and MobileNet(x_i) across k categories are denoted as:

$$\text{VGG16}(x_i) = [p_{1i}(x_i), p_{2i}(x_i), \ldots, p_{ki}(x_i)] \tag{2}$$

$$\text{MobileNet}(x_i) = [q_{1i}(x_i), q_{2i}(x_i), \ldots, q_{ki}(x_i)] \tag{3}$$

Where, $p_{ji}(x_i)$ and $q_{ji}(x_i)$ are the probabilities assigned to a specific category $j(1 \leq j \leq k)$ by model VGG16 and MobileNet respectively.

2. Combining the predictions of VGG-16 and MobileNet into a stacked feature vector for each instance i, we get the final ensembled prediction, E_{ij} of a specific category j:

$$E_{ij} = \frac{1}{2} \sum_{n=1}^{2} M_n(x_i) \tag{4}$$

where $M_n(x_i)$ is probability distribution given by Model M_n for instance x_i

3. For k categories, the category j with the highest probability will be predicted as the final predicted class $Y_{\text{final},i}$ for image x_i as below:

$$Y_{\text{final},i} = \text{argmax}_j \left(E_{ij}\right) \tag{5}$$

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

4.2 Recommendation

For the product recommendation module, The input images are fed into the feature extraction sub-module where the bagging ensemble model which is a combination of MobileNet and RetNet5o is used for image pre-processing and feature extraction from the query image. Then this processed image is fed to the image similarity module where KNN and Euclidean distance will be used to calculate the distance of all the images from the query image. Finally, the result comes as a set of similar images to the query images. That is for Nectarine as a query image, the images of other similar Fruits like Nectarine, Apple and Pomegranate come as the recommended images.

This is how the whole architecture facilitates the search result of that query and recommends similar products that the users might like.

5 Experimental Result

In this section, we present the results of the evaluation for aforementioned the visual searching and product recommendation techniques.

5.1 Performance Evaluation of Visual Searching Method

5.1.1 Evaluating Criteria

Our performance evaluation incorporates widely used evaluation metrics such as accuracy, precision, recall and F1 score. The performance metrics are based on a comparative analysis of predicted values and true values of the dataset which can be categorized into four categories: true positive where both the true value and predicted value are positive; true negative where both the true value and predicted value are negative; false positive where the true value is negative but the predicted value resulted to be positive; false negative in which actual value is positive but predicted to be negative. The performance matrices can be written as follows:

$$Accuracy = \frac{True\ Positives + True\ Negatives}{True\ Positives + True\ Negatives + False\ Positives + False\ Negatives} \tag{6}$$

$$Precision = \frac{True Positives}{True\ Positives + False\ Positives} \tag{7}$$

$$Recall = \frac{True\ Positives}{True\ Positives + False\ Negatives} \tag{8}$$

$$F1\,score = \frac{2 \cdot precision \cdot recall}{precision + recall} \tag{9}$$

5.1.2 Performance Evaluation of Deep Learning Techniques

Table 2 shows the performance comparison of the three state-of-the-art models that have been employed for technique 1. The time to get results from a machine learn-

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

Table 2 Performance analysis of collected test data from Kaggle using traditional machine learning technique (Technique 1)

Models	Accuracy	Precision	Recall	F1	Computation time
VGG16	94%	85%	70%	80%	2 min 15 s
MobileNet	**99.21%**	**99%**	**98%**	**98%**	**13 s**
ResNet50	99%	80%	82%	73%	43 s

The highest percentage of accuracy, precision, recall, F1 and the shortest computation time

Table 3 Performance analysis of prepared test dataset using traditional machine learning technique (Technique 2)

Models	Accuracy	Precision	Recall	F1	Computation time
VGG16	90.625%	83%	75%	70%	35 s
MobileNet	**98.43%**	**99%**	**98%**	**98%**	**17 s**
ResNet50	98.43%	78%	83%	65%	33 s

The highest percentage of accuracy, precision, recall, F1 and the shortest computation time

ing model is an important consideration that can impact the usability, effectiveness, and efficiency of the model in various applications. It's shown from the table that, MobileNet is giving the highest accuracy and lowest computation time while VGG16 performs the worst among the three. MobileNet also gives the highest precision, recall and F1 score among all the models.

Technique 2 also deals with traditional machine learning models which use the prepared data set. The performance metrics are kept similar here. Based on the data presented in Table 3, both MobileNet and ResNet50 exhibit identical accuracy levels of 98.43% but MobileNet requires less time than other models. Again MobileNet gives 99% precision while ResNet50 gives 78% precision. This difference is also seen in the case of recall and F1-score. Thus, it is clear that MobileNet performs better than the other two algorithms for both data sets. Thus, it is clear that MobileNet outperforms other models for both the collected data set from Kaggle (Technique 1) and the self-prepared data set (Technique 2).

5.1.3 Performance Analysis of Ensemble Learning Techniques

Tables 4 and 5 show the Accuracy, Precision, F1 score, and Computation Time for all possible combinations of the machine learning models that is the ensemble learning for Technique 3 and Technique 4 respectively. According to Table 4, all the combination demonstrates the varying levels of accuracy for the collected data set from Kaggle with MobileNet and VGG16 having the highest 99.78% accuracy. It also has the highest precision, recall and F1 score among all the combinations but needs slightly more execution time than ResNet50 and MobileNet combinations. ResNet50 and MobileNet show 97.2% accuracy and the computation time, precision, recall and F1 score is slightly less than that of MobileNet and VGG16 combination. The combination of all three models also shows good accuracy but requires high computation time.

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

Table 4 Performance analysis of collected test data from Kaggle using ensemble learning learning technique (Technique 3)

Combination	Accuracy	Precision	Recall	F1	Computation time
VGG16 and ResNet50	80.4%	63%	60%	59%	3 min 10 s
MobileNet and VGG16	**99.78%**	**99%**	**99%**	**98%**	3 min 48 s
ResNet50 and MobileNet	97.2%	92%	93%	90%	**2 min 4 s**
VGG16, ResNet50, MobileNet	96.8%	90%	88%	85%	3 min 57 s

The highest percentage of accuracy, precision, recall, F1 and the shortest computation time

Table 5 Performance analysis of prepared test data set using ensemble learning technique (Technique 4)

Combination	Accuracy	Precision	Recall	F1	Computation time
VGG16 and ResNet50	89.9%	70%	78%	75%	1 min 7 s
MobileNet and VGG16	**99.8%**	**99%**	**97%**	**98%**	**49 s**
ResNet50 and MobileNet	98%	85%	83%	70%	1 min 9 s
VGG16, ResNet50, MobileNet	98%	90%	86%	89%	1 min

The highest percentage of accuracy, precision, recall, F1 and the shortest computation time

The performance analysis of the ensemble learning technique with a self-prepared data set is demonstrated in Table 5. It is evident from the table that the combination of MobileNet and VG16 gives an outstanding performance just as the collected data set with a lesser computation time of 49 s. The other combinations also perform better and take less computation time than that of the Kaggle data set (Technique 3). As MobileNet and VGG16 outperform in terms of accuracy and also computation time for both data sets it is chosen in the proposed model for product classification.

5.2 Product Recommendation

For each model, the output of similar recommended images is manually analyzed for a given query image. The evaluation is based on accuracy which helps to assess the effectiveness of recommended images generated by models. Here moderate accuracy is considered instead of greater accuracy as recommended images are expected to be a broader range of that class having the query product and similar products to it.

5.2.1 Performance Analysis of Deep Learning Techniques

An image of Milk from the Packages class is given as input and the resultant recommended images for each model are observed in Fig. 7. All the models give outputs of different types of products from the packages class. From Fig. 7a, it is seen that the combination of VGG16 and KNN gives images of four milk, two beverages, one oat milk, one sour milk and one sour cream. The combination of MobileNet and KNN (Fig. 7b) gives four milk, two beverages, two soy milk and one sour cream image.

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

(a) Recommended Images: VGG-16 & KNN

(b) Recommended Images: MobileNet & KNN

(c) Recommended Images: ResNet-50 & KNN

Fig. 7 Results of product recommendation using deep learning techniques

Lastly, the combination of ResNet50 and KNN (Fig. 7c) gives four milk images and five beverage images.

5.2.2 Performance Analysis of Ensemble Learning Techniques

The same Milk input image used in deep learning techniques is given as input here also. The recommended images of all the combinations are observed in Fig. 8. The combination of ensemble model 1(VGG16 and ResNet50) and KNN gives four milk images and five beverage images (Fig. 8a). Figure 8b shows the combination of ensemble model 2 (MobileNet and VGG16) and KNN which gives seven images of milk. Five images of milk and two images of soy milk are given by the combination of ensemble model 3 (ResNet50 and MobileNet) and KNN and it is depicted in Fig. 8c.

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

(a) Recommended Images: Ensemble Model 1 & KNN

(b) Recommended Images: Ensemble Model 2 & KNN

(c) Recommended Images: Ensemble Model 3 & KNN

(d) Recommended Images: Ensemble Model 4 & KNN

Fig. 8 Results of product recommendation using ensemble-learning techniques

Lastly, the combination of ensemble model 4 (VGG16, ResNet50 and MobileNet) and KNN gives five milk, two soy milk, one sour cream, and one yoghurt image as depicted in Fig. 8d.

5.3 Comparative Analysis

5.3.1 Visual Search

The comparative performance analysis of the accuracy and execution times of various machine learning models employing the four types of techniques in this study for the visual search of products are represented graphically in Figs. 9 and 10 respectively.

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

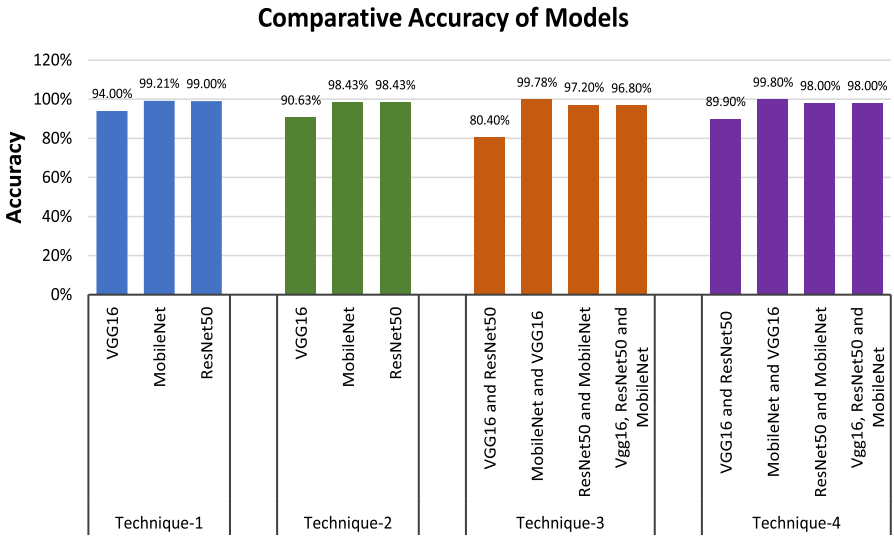

Fig. 9 Comparative analysis of accuracy using different techniques

The comparative analysis of the four techniques for getting search results using two types of data sets reveals that Technique 4 which is the Ensemble learning technique with the prepared data set performs the best. The deep learning and ensemble learning technique with the Kaggle data set gives a very high execution time which is not preferable. Among both deep learning techniques, it is evident that the MobileNet model performs the best among the other three models with approx 99% accuracy and 15sec execution time. The combination of MobileNet and VGG16 performs the best with a 99.79% accuracy on average among all the combinations used in the ensemble learning techniques. Now if these two best models from the two types of techniques are to be considered, then the ensemble learning model which is the combination of MobileNet and VGG16 gives the most desirable result.

In summary, as the prepared data set had different varieties of images, the focus was to achieve better results using this data set which seems to be fulfilled. Again as the ensemble learning technique performs better than the deep learning techniques therefore ensemble learning approach is recommended to implement the visual search of products in the proposed model of this research.

5.3.2 Product Recommendation

Recommended images are expected to be a generalized class of the query product image. That better accuracy is not desirable here. For example, if an image of Milk is given as input then similar recommended images will be other packaged groceries along with different milk images. Better accuracy models will give accurate results of milk only. That is why the models that performed the best in the visual search part are seen to have given only milk images as output. Among the other models, it is noticed that ResNet50 from the deep learning technique and Ensemble Model 1 from

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

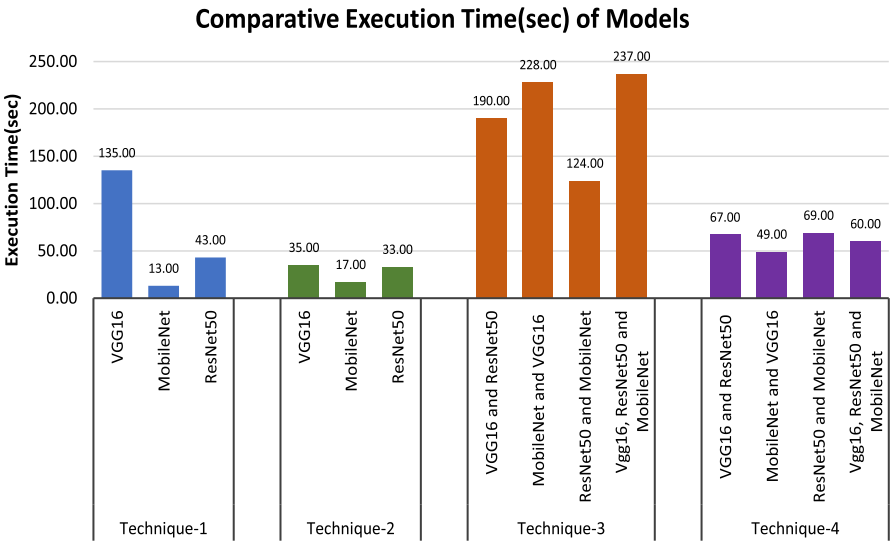

Fig. 10 Comparative analysis of execution time using different techniques

Table 6 5-fold cross-validation results

Test set	Accuracy	Precision	Recall	F1 score
1	93.94%	94.12%	93.93%	93.76%
2	96.63%	96.29%	96.30%	96.28%
3	96.97%	96.97%	96.97%	96.96%
4	95.30%	96.37%	96.30%	96.23%
5	97.64%	97.66%	97.64%	97.63%
Average	96.23%	96.28%	96.22%	96.18%

the ensemble learning technique recommend too many images of beverages rather than milk. Therefore these models are also not recommended. Now from the three models left, VGG16 and MobileNet of deep learning technique are seen to have a noticeable amount of beverage images. Thereby, it is evident that ensemble model 4 which is the ResNet50 and MobileNet combination with KNN is the best fit because it recommended similar categories of milk that are sour milk, soy milk, soy cream etc.

5.4 K-fold Cross-Validation

The prepared data set was split into five non-overlapping sections to do cross-validation. The proposed model (The combination of VGG-16 and Mobilenet) has been trained five times for the five sections each. Moreover, the accuracy, precision, recall and f1-score of each section were calculated and shown in Table 6. In addition, the average for the whole process has been mentioned at the end.

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

6 Discussion and Conclusion

Using deep learning models and ensemble learning approaches, this research aims to create a system for visual search and recommendation of grocery products. A system is also proposed, that offers comparable images to users based on their surfing behaviour. The system's image search component scored a high accuracy rate, with an F1-score of 0.98 and an average precision of 0.98. When compared to one deep learning model, the model's performance was found to be significantly improved by the ensemble learning strategy. The recommendation system performed admirably as well. The study has shown that deep learning models and ensemble learning strategies are effective for visual search and comparable image recommendation. Users can engage with the system and verify its functionality in real time using the website that is being developed. The finding has important practical ramifications for a variety of applications, including image retrieval for research and education as well as e-commerce.

The fact that the visual search and recommendation system is only trained and assessed on a grocery data set is one of the critical limitations of this study. The scope of this data set is still constrained in comparison to the full range of visual objects and categories, even though it contains a lot of grocery products and poses a considerable challenge for picture retrieval and recommendation. The future work of this research can be expanding the data set even further. Currently, the data set only has three major categories: Fruits, Vegetables, and Packages. The goal is to add more categories that exist in a real grocery shop to make the proposed system more convenient. The current data set contains images that are blurred or captured in low light along with clear images of the products. In the future, the system will have a better pre-processing step to prepare the pictures even better for the models that are used in the proposed system.

Funding The authors did not receive support from any organization for the submitted work.

Data availability Both of the prepared dataset and Kaggle dataset are openly available in GitHub at https://github.com/The-Talking-Head/Sourcecode- and-Dataset, under the folder of Dataset and Kaggle Dataset respectively.

Code Availability The codes for this study are openly available in GitHub at https://github.com/The-Talking-Head/Source-code-and-Dataset, under the folder of Source Code.

Declarations

Conflict of interest All authors certify that they have no affiliations with or involvement in any organization or entity with any financial interest or non-financial interest in the subject matter or materials discussed in this manuscript.

References

1. Imran (2020) On-site search statistics for ecommerce and others. https://www.addsearch.com/blog/site-search-statistics. Accessed 26 June 2023

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

2. pixyle.ai (2020) Is visual search making text-based search obsolete? https://www.pixyle.ai/product-discovery/is-visual-search-making-text-based-search-obsolete. Accessed 28 July 2023
3. Shi Y (2022) Advances in big data analytics: theory, algorithm and practice. Springer, Singapore
4. Olson DL, Shi Y (2007) Introduction to business data mining. McGraw-Hill/Irwin, New York
5. Shi Y, Tian YJ, Kou G, Peng Y, Li JP (2011) Optimization based data mining: theory and applications. Springer, Berlin
6. Tien JM (2017) Internet of things, real-time decision making, and artificial intelligence. Ann Data Sci 4(2):149–178
7. Felfernig A, Jeran M, Ninaus G, Reinfrank F and Reiterer S, Stettinger M (2014) Basic approaches in recommendation systems
8. Umer S, Mohanta PP, Rout RK, Pandey HM (2021) Machine learning method for cosmetic product recognition: a visual searching approach. Multimed Tools Appl 80:34997–35023. https://doi.org/10.1007/s11042-020-09079-y
9. Bishop CM, Nasrabadi NM (2006) Pattern recognition and machine learning. Springer, New York
10. Chang CC, Lin CJ (2011) A library for Support Vector Machines. ACM Trans Intell Syst Technol 2(3):2157–6904. https://doi.org/10.1145/1961189.1961199
11. Mullick SS, Datta S, Das S (2018) Adaptive learning-based k-nearest neighbor classifiers with resilience to class imbalance. IEEE Trans Neural Netw Learn Syst 29(11):5713–5725. https://doi.org/10.1109/TNNLS.2018.2812279
12. Raschka S (2015) Python machine learning. Packt Publishing Ltd, Birmingham
13. Safavian SR, Landgrebe D (1991) A survey of decision tree classifier methodology. IEEE Trans Syst Man Cybern 21(3):660–674. https://doi.org/10.1109/21.97458
14. Rinaldi AM, Russo C, Tommasino C (2021) Visual query posing in multimedia web document retrieval. In: Proceedings of the 15th international conference on semantic computing (ICSC). IEEE, Laguna Hills. https://doi.org/10.1109/ICSC50631.2021.00086
15. Smelyakov K, Sandrkin D, Ruban I, Martovytskyi V, Romanenkov Y (2018) Search by image. New search engine service model. In: Proceedings of the international scientific-practical conference problems of infocommunications. Science and Technology (PIC S&T). IEEE, Kharkiv. https://doi.org/10.1109/INFOCOMMST.2018.8632117
16. Hu C, Li Q, Zhang Z, Chang Kh, Zhang R (2020) A multimodal fusion framework for brand recognition from product image and context. In: Proceedings of the international conference on multimedia & expo workshops (ICMEW), pp 1–4. https://doi.org/10.1109/ICMEW46912.2020.9105947
17. Wei Z, Song X and Lin Y (2020) Image feature recognition algorithm for rural revitalization product design based on visual attention model. In: Proceedings of the 4th international conference on inventive systems and control (ICISC). IEEE, Coimbatore. https://doi.org/10.1109/ICISC47916.2020.9171207
18. Sogi N, Souza LS, Gatto BB, Fukui K (2020) Metric learning with a-based scalar product for image-set recognition. In: Proceedings of the conference on computer vision and pattern recognition workshops (CVPRW). IEEE, Seattle. https://doi.org/10.1109/CVPRW50498.2020.00433
19. Boriya A, Malla SS, Manjunath R, Velicheti V, Eirinaki, M (2019) ViSeR: A visual search engine for e-retail. In: Proceedings of the first international conference on transdisciplinary AI (TransAI). IEEE, Laguna Hills. https://doi.org/10.1109/TransAI46475.2019.00021
20. Dakhare BS, Khatu A, Singh HR, Gupta A (2020) Visual E-commerce application using deep learning. Int Res J Eng Technol (IRJET) 7(3):1–9
21. Kim Y (2021) A study on the efficient product search service for the damaged image information. arXiv preprint arXiv:2111.07346
22. Zhang Ya, Pan P, Zheng Y, Zhao K, Zhang Y, Ren X and Jin R (2018) Visual search at Alibaba. In: Proceedings of the 24th ACM SIGKDD international conference on knowledge discovery & data mining, pp 993–1001. https://doi.org/10.1145/3219819.3219820
23. Szegedy C, Wei L, Yangqing J, Sermanet P, Reed S, Anguelov D, Erhan, Vanhoucke V, Rabinovich A (2015) Going deeper with convolutions. In: Proceedings of the IEEE conference on computer vision and pattern recognition (CVPR), pp 1–9. https://doi.org/10.1109/CVPR.2015.7298594
24. Russakovsky O, Deng J, Su H, Krause J, Satheesh S, Ma S, Huang Z, Karpathy A, Khosla A, Bernstein M, Berg A, Fei-Fei L (2015) Imagenet large scale visual recognition challenge. Int J Comput Vis 115:211–252. https://doi.org/10.1007/s11263-015-0816-y
25. Li J, Liu H, Gui C, Chen J, Ni Z, Wang N, Chen Y (2018) The Design and implementation of a real time visual search system on JD E-commerce platform. In: Proceedings of the 19th international middleware conference industry. https://doi.org/10.1145/3284028.3284030

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

26. Hu H, Wang Y, Yang L, Komlev P, Huang L, Chen X, Huang J, Wu Y, Merchant M, Sacheti A (2018) Web-scale responsive visual search at bing. In: Proceedings of the 24th ACM SIGKDD international conference on knowledge discovery & data mining, pp 359–367. https://doi.org/10.1145/3219819.3219843
27. Smelyakov K, Chupryna A, Sandrkin D, Kolisnyk M (2020) Search by image engine for big data warehouse. In: Proceedings of the 2020 IEEE open conference of electrical, electronic and information sciences (eStream), pp 1–4
28. Mu C, Zhao J, Yang G, Zhang J, Yan Z (2018) Towards practical visual search engine within elasticsearch. arXiv preprint arXiv:1806.08896
29. Karmakar C, Datcu M (2020) A fast search system for remote sensing imagery based on bag of visual words and latent Dirichlet allocation. In: Proceeding of the international geoscience and remote sensing symposium (IGARSS). IEEE, Waikoloa. https://doi.org/10.1109/IGARSS39084.2020.9324359
30. Krizhevsky A, Sutskever I, Hinton GE (2017) ImageNet classification with deep convolutional neural networks. Commun ACM 60:84–90. https://doi.org/10.1145/3065386
31. He K, Zhang X, Ren A, Sun J (2015) Spatial pyramid pooling in deep convolutional networks for visual recognition. IEEE Trans Pattern Anal Mach Intell 37:1904–1916. https://doi.org/10.1109/TPAMI.2015.2389824
32. Ioffe S, Szegedy C (2015) Batch normalization: accelerating deep network training by reducing internal covariate shift. arXiv. https://doi.org/10.48550/arXiv.1502.03167
33. He K, Zhang X, Ren S, Sun J (2016) Deep residual learning for image recognition. In: Proceedings of the IEEE conference on computer vision and pattern recognition (CVPR). IEEE, Las Vegas. https://doi.org/10.1109/CVPR.2016.90
34. Marichal X, De VC, Macq B (1997) Towards visual search engine based on fuzzy logic
35. Sumaiya, Armanuzzaman M (2020) Enhancement of resulting image search engine (ERISE) by content-based image retrieval system. In: Proceedings of the IEEE region 10 symposium (TENSYMP). IEEE, Dhaka. https://doi.org/10.1109/TENSYMP50017.2020.9230653
36. Simonyan K, Zisserman A (2015) Very deep convolutional networks for large-scale image recognition. arXiv. https://doi.org/10.48550/arXiv.1409.1556
37. Hur C, Hyun C, Park H (2020) Automatic image recommendation for economic topics using visual and semantic information. In: Proceedings of the 14th international conference on semantic computing (ICSC). IEEE, San Diego. https://doi.org/10.1109/ICSC.2020.00037
38. Redmon J, Farhadi A (2017) YOLO9000: better, faster, stronger. In: Proceedings of the IEEE conference on computer vision and pattern recognition (CVPR). IEEE, Honolulu. https://doi.org/10.1109/CVPR.2017.690
39. Tautkute I, Trzciński T, Skorup AP, Brocki L, Marasek K (2019) DeepStyle: multimodal search engine for fashion and interior design. IEEE Access 7:84613–84628. https://doi.org/10.1109/ACCESS.2019.2923552
40. Eswaran A, Varshini E (2022) Reverse image search engine for garment industry. In: Proceedings of the 2022 8th international conference on advanced computing and communication systems (ICACCS). IEEE, Coimbatore, pp. 414–418

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

Terms and Conditions

Springer Nature journal content, brought to you courtesy of Springer Nature Customer Service Center GmbH ("Springer Nature").

Springer Nature supports a reasonable amount of sharing of research papers by authors, subscribers and authorised users ("Users"), for small-scale personal, non-commercial use provided that all copyright, trade and service marks and other proprietary notices are maintained. By accessing, sharing, receiving or otherwise using the Springer Nature journal content you agree to these terms of use ("Terms"). For these purposes, Springer Nature considers academic use (by researchers and students) to be non-commercial.

These Terms are supplementary and will apply in addition to any applicable website terms and conditions, a relevant site licence or a personal subscription. These Terms will prevail over any conflict or ambiguity with regards to the relevant terms, a site licence or a personal subscription (to the extent of the conflict or ambiguity only). For Creative Commons-licensed articles, the terms of the Creative Commons license used will apply.

We collect and use personal data to provide access to the Springer Nature journal content. We may also use these personal data internally within ResearchGate and Springer Nature and as agreed share it, in an anonymised way, for purposes of tracking, analysis and reporting. We will not otherwise disclose your personal data outside the ResearchGate or the Springer Nature group of companies unless we have your permission as detailed in the Privacy Policy.

While Users may use the Springer Nature journal content for small scale, personal non-commercial use, it is important to note that Users may not:

1. use such content for the purpose of providing other users with access on a regular or large scale basis or as a means to circumvent access control;
2. use such content where to do so would be considered a criminal or statutory offence in any jurisdiction, or gives rise to civil liability, or is otherwise unlawful;
3. falsely or misleadingly imply or suggest endorsement, approval , sponsorship, or association unless explicitly agreed to by Springer Nature in writing;
4. use bots or other automated methods to access the content or redirect messages
5. override any security feature or exclusionary protocol; or
6. share the content in order to create substitute for Springer Nature products or services or a systematic database of Springer Nature journal content.

In line with the restriction against commercial use, Springer Nature does not permit the creation of a product or service that creates revenue, royalties, rent or income from our content or its inclusion as part of a paid for service or for other commercial gain. Springer Nature journal content cannot be used for inter-library loans and librarians may not upload Springer Nature journal content on a large scale into their, or any other, institutional repository.

These terms of use are reviewed regularly and may be amended at any time. Springer Nature is not obligated to publish any information or content on this website and may remove it or features or functionality at our sole discretion, at any time with or without notice. Springer Nature may revoke this licence to you at any time and remove access to any copies of the Springer Nature journal content which have been saved.

To the fullest extent permitted by law, Springer Nature makes no warranties, representations or guarantees to Users, either express or implied with respect to the Springer nature journal content and all parties disclaim and waive any implied warranties or warranties imposed by law, including merchantability or fitness for any particular purpose.

Please note that these rights do not automatically extend to content, data or other material published by Springer Nature that may be licensed from third parties.

If you would like to use or distribute our Springer Nature journal content to a wider audience or on a regular basis or in any other manner not expressly permitted by these Terms, please contact Springer Nature at

onlineservice@springernature.com